

export air quality data and then give a name to your exported file. BenMAP-CE will generate a text file that you can then examine. This is discussed in detail in Chapter 3, Section 3.2.1 (Tools).

How do I perform a rollback to simulate attainment with an annual *and* daily PM_{2.5} standard?

Unfortunately, BenMAP-CE will roll back to either an annual standard or a daily standard—but not both. If this feature is of interest to you, please contact the BenMAP-CE developers at benmap@epa.gov.

Chapter 6

Estimating Incidence

In this chapter...

- **Get an overview of how BenMAP-CE estimates the incidence of health outcomes.**
- **Learn to create a health impact configuration.**
- **Learn about baseline and control scenarios.**
- **Learn the difference between Point Mode and the Monte Carlo analysis options.**
- **Learn how to run, save, and re-open a configuration.**
- **Learn how to view and export Incidence Results.**

To estimate changes in the incidence (or, counts) of adverse health effects from air pollution, you will need to create and run a BenMAP-CE configuration file (*.cfgx). A configuration is a reusable file that specifies the air quality grids, health impact functions, population data, and other parameters necessary for an analysis. It is a record of the choices you make in estimating the change in adverse health effects between a baseline and control scenario. The choices include the following:

- The pollutant and air quality grids for the baseline and control scenarios;
- The year for the analysis;
- The population dataset for the analysis;
- The health impact functions to be used in estimating adverse health effects; and Whether the analysis will focus on a single "point" estimate (Point Mode), or a range of results that mirror the variability in the inputs to the health impact functions (Monte Carlo-generated percentiles).

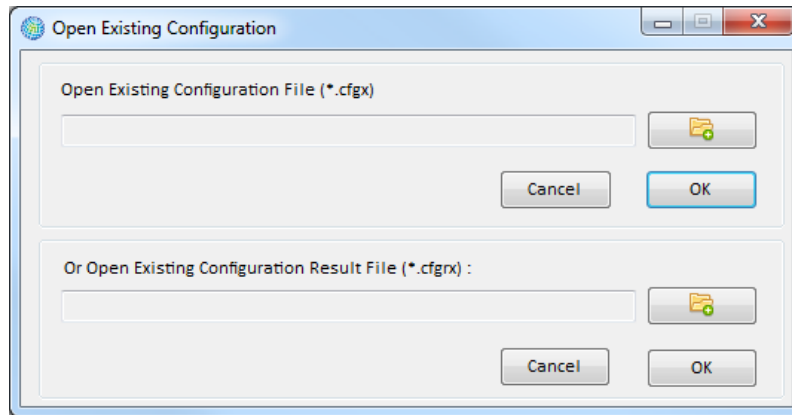
Fundamental Concept – Configuration Files

A **Health Impact Configuration** file (*.cfgx) includes all the user-specified data and choices for a BenMAP-CE run made prior to executing the health impact function step. This includes specified air quality grids, air quality surfaces, population data, health impact functions, and incidence datasets. This file is not saved automatically but can be saved by clicking the appropriate button. Configuration files can be useful for running BenMAP-CE in batch mode via the Command Line feature (see Appendix L).

Once you execute the Health Impact Functions, BenMAP will save a **Health Impact Configuration Results** file (*.cfgrx). This file contains the same information as the .cfgx file, plus the health impacts results from running the health impact functions on the specified air quality surfaces.

BenMAP-CE gives you flexibility in creating, editing, and saving configuration files. You can open an already existing configuration and proceed directly to estimate incidence. Or, you can create a new file, and then estimate incidence. In addition, you may save any edits made to existing configuration files. After calculating the change in adverse health effects, BenMAP-CE saves the results in a "configuration results" file with a *.cfgrx* extension. The results obtained from running a configuration are sometimes referred to as "raw" results because they represent the estimated change in incidence for each air quality grid cell in a given scenario; they have not been aggregated, pooled, or valued (see Chapter 7).

To load a previously saved configuration, click **Estimate Health Impacts**, which is beside **Step 2** in the BenMAP-CE tree menu (on the left side of the main window). An **Open Existing Configuration** window will appear, as shown below:



To open an existing Configuration file, click the top file browser button. To open an existing Configuration Result file, click the bottom file browser button. After selecting either of these options, search and select a file to load in the **Open** window. The file path and name should appear in the box next to the file browser button. Once you have selected the desired option, and are satisfied with the file selection, click **OK**.

6.1 Introduction to Estimating Health Incidence

Health impact functions relate a change in the concentration of a pollutant to a change in the incidence of a health endpoint (i.e., premature mortality or work-loss days). It is typically derived from the estimated relationship between the concentration of a pollutant and the adverse health effects suffered by a given population in an epidemiology study. For example, the pollutant concentration being measured may be particulate matter (PM_{2.5}), and the population response may be daily premature deaths. For the purposes of estimating benefits, we are not interested in the health impact function itself, but rather the relationship between the change in concentration of the pollutant, and the corresponding change in the population-health response. We may want to know, for example, if the concentration of PM_{2.5} is reduced by 10µg/m³, how many premature deaths will be avoided?

To estimate changes in health incidence, the first step is to calculate the change in pollution

concentrations for a particular policy scenario, such as an air quality improvement produced by a set of emissions controls. The concentration change in a pollutant is the

Fundamental Concept: Health Impact Function or Concentration-Response Function

A **health impact function** calculates the change in the number of adverse health effects associated with a change in exposure to air pollution. The inputs to a health impact function include the change in air quality concentration for a pollutant (using a specified metric such as annual D24HourMean); the size of the affected population (of specified age, race and ethnicity); the baseline incidence rate of the adverse health effect; and an effect coefficient derived from epidemiological studies.

The coefficient for the health impact function is known as Beta (β) and is derived from epidemiological studies. The value of β typically represents the percent change in a given adverse health impact per unit of pollution.

Health impact functions are derived from **concentration-response (C-R) functions**, which estimate the relationship between the likelihood of adverse health effects as a function of concentration of an air pollutant. The terms C-R function and health impact function are often used interchangeably.